

CACERES, Spain

WMO: 082610

Lat: 39.4714N

Lon: 6.3389W

Elev: 1283

StdP: 14.03

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.1	34.6	21.9	17.2	43.8	24.9	19.9	43.7	20.5	51.2	18.6	51.0	2.5	230	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.6	99.6	67.3	97.1	66.8	94.3	66.1	70.1	93.1	68.7	91.4	67.5	89.4	6.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.2	90.8	74.4	61.8	86.6	72.7	60.6	82.8	71.6	34.7	93.1	33.5	91.3	32.5	89.7	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.1	15.9	14.1	DB	27.7	104.7	1.9	2.9	26.3	106.8	25.2	108.5	24.1	110.2	22.7	112.3
			WB	25.9	74.1	2.2	3.7	24.4	76.7	23.1	78.9	21.9	80.9	20.3	83.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.1	46.7	49.1	54.1	58.0	64.8	74.8	79.6	79.3	72.1	64.1	53.7	48.1	(6)
	DBStd	12.96	4.80	4.75	5.42	6.07	6.71	7.17	5.78	5.40	5.81	5.35	4.97	4.83	(7)
	HDD50	333	124	68	23	5	0	0	0	0	0	0	19	94	(8)
	HDD65	2625	568	445	339	225	89	8	0	0	7	80	341	523	(9)
	CDD50	4757	21	43	150	245	460	744	919	909	664	437	129	36	(10)
	CDD65	1572	0	0	0	14	84	303	454	444	221	51	1	0	(11)
	CDH74	18686	0	0	10	178	918	3593	5854	5592	2167	374	0	0	(12)
	CDH80	9366	0	0	0	22	296	1794	3244	3017	895	98	0	0	(13)
(14) Wind	WSAvg	6.2	5.9	6.5	7.0	6.8	6.5	6.4	6.5	6.1	5.9	5.7	6.0	5.4	(14)
(15) Precipitation	PrecAvg	21.5	2.3	2.0	1.9	2.0	1.8	0.6	0.2	0.2	1.3	3.5	3.1	2.8	(15)
	PrecMax	37.9	9.6	6.6	8.6	5.8	4.2	2.3	1.4	1.0	4.6	8.7	14.0	7.9	(16)
	PrecMin	12.6	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.3	0.0	0.1	(17)
	PrecStd	6.1	2.0	1.6	1.9	1.3	1.3	0.6	0.3	0.3	1.0	2.0	2.9	2.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.5	65.8	75.3	82.8	91.5	100.6	102.8	103.2	98.1	87.7	71.7	62.5	(19)
		MCWB	52.9	53.8	56.4	62.0	63.6	68.3	67.6	68.4	65.2	61.3	59.8	55.2	(20)
	2%	DB	58.2	62.4	71.6	79.1	86.9	96.7	99.9	99.2	92.3	82.2	67.3	59.7	(21)
		MCWB	52.7	52.7	55.5	59.6	62.5	67.5	67.0	67.5	65.8	60.2	57.7	55.0	(22)
	5%	DB	56.2	59.7	68.5	75.0	82.8	93.5	97.3	96.5	88.4	77.7	64.3	57.7	(23)
		MCWB	51.6	51.7	54.9	58.5	61.2	66.1	66.4	66.8	63.9	61.0	56.6	53.6	(24)
	10%	DB	54.3	57.2	65.1	70.3	79.2	90.2	94.4	93.5	85.0	73.9	61.7	55.6	(25)
		MCWB	50.4	51.1	53.7	56.2	60.2	65.4	65.9	66.0	63.1	59.8	56.0	52.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.6	57.3	59.3	64.7	67.2	71.6	71.6	72.9	69.6	66.4	62.7	58.2	(27)
		MCDB	58.9	61.3	70.3	79.4	84.9	94.6	94.7	92.8	88.0	77.5	66.7	59.9	(28)
	2%	WB	54.3	55.2	57.5	61.6	64.5	69.2	69.7	70.2	67.4	64.6	60.4	56.3	(29)
		MCDB	56.6	59.2	68.2	75.8	81.6	91.5	94.2	93.1	86.5	73.4	64.4	58.7	(30)
	5%	WB	52.6	53.3	56.0	59.3	62.8	67.6	68.2	68.7	66.1	63.3	58.5	54.3	(31)
		MCDB	55.2	57.6	65.5	72.4	78.9	89.4	92.4	91.3	83.3	71.4	62.6	57.0	(32)
	10%	WB	51.0	51.7	54.4	57.2	61.2	66.2	66.9	67.3	64.8	62.0	56.8	52.3	(33)
		MCDB	53.8	56.7	63.2	68.2	75.7	87.1	90.1	89.2	80.2	69.8	60.7	55.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	14.8	17.9	18.4	19.8	22.6	24.6	24.4	21.0	17.0	13.9	13.0	(35)
		MCDBR	15.7	18.3	23.9	25.5	26.0	26.2	27.1	27.2	26.5	24.0	18.3	14.2	(36)
	5% WB	MCWBR	10.6	10.7	11.9	11.6	9.6	8.7	8.4	9.2	9.8	10.0	10.1	9.8	(37)
		MCWBR	11.3	14.0	20.6	22.6	23.1	24.5	25.3	24.0	21.8	16.4	12.9	11.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.303	0.318	0.351	0.357	0.366	0.368	0.350	0.363	0.356	0.341	0.316	0.302	(40)
		taud	2.504	2.466	2.388	2.382	2.384	2.395	2.412	2.381	2.423	2.490	2.512	2.513	(41)
	Ebn at Noon		277	287	287	291	289	288	292	285	282	275	269	268	(42)
		Edh at Noon	24	29	35	37	38	37	36	37	33	28	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		707	1021	1375	1746	2125	2416	2507	2186	1676	1127	780	625	(44)
	RadStd		102	133	163	167	172	122	62	70	101	106	105	78	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+117

Nomenclature: See separate page